Edward Ajayi

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https://edwardajayi.github.io/

Education

Carnegie Mellon University Rwanda
Master of Science in Engineering Artificial Intelligence *Aug. 2024 – May. 2026*
- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.73/4.0

Ekiti State University Nigeria
Bachelor of Science in Computer Science *Oct. 2014 – Nov. 2019*
- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Human-Centered AI, Socially Aware NLP, Computational Pragmatics, Language Modeling, AI Alignment & Evaluation

Selected Publications

1. **E. Ajayi** and P. Mitra. *HumorGen: Cognitive Synergy for LLM Humor Generation via Persona-Based Distillation. Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main Conference), to appear.* [arXiv Preprint]
2. **E. Ajayi** and P. Mitra. *Cross-Lingual Cognitive Synergy for Constrained Humor Generation in LLMs.* **Accepted to *Conference and Labs of the Evaluation Forum (CLEF) 2026 JOKER Track***. 2026. [Manuscript]
3. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR), AAAI-26 AI in Medicine and Healthcare Bridge.* **Best Paper Award**. 2026. [PMLR]
4. **E. Ajayi** and P. Mitra. *HumorRank: An Automated Leaderboard for Evaluating Humor Generation in Large Language Models.* Submitted to *AAAI*, 2027. [arXiv Preprint]
5. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages. Deep Learning Indaba 2026 Research Track, PMLR, to appear.* [arXiv Preprint]
6. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models.* Under Review. 2026. [Manuscript]
7. **E. Ajayi**. *AfriEconQA: A Benchmark Dataset for African Economic Analysis Based on World Bank Reports.* Under Review. 2026. [arXiv Preprint]

Experience

Research Associate
Spatial and Language Technologies Lab, Carnegie Mellon University *Jun. 2026 – Sept. 2026*
- Building efficient multimodal humor generation models using LLMs and vision-language models (VLMs).
- Developing evaluation benchmarks for Large Language Models on subjective generation tasks and alignment.
- Conducting research on computational humor, multilingual NLP, and reasoning abilities of foundation models.

Predoctoral Research Fellow
Human-Algorithm Interaction Lab, Oxford Saïd Business School, University of Oxford, UK *Feb. 2026 – May. 2026*
- Conduct research on Human-AI collaboration, decision-making, and AI adoption in service systems at the Human-Algorithm Interaction Lab.

Research Assistant
Spatial and Language Technologies Lab, Carnegie Mellon University *Aug. 2025 – May. 2026*
- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model training, evaluation, and benchmarking.
- Contributing to multilingual and low-resource language technologies focused on culturally grounded language models.

Research Intern - Business Automation
Carnegie Mellon University *Jun. 2025 – Aug. 2025*
- Built an agentic AI workflow framework using LangGraph and FastAPI to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Technical Skills

Languages: Python, SQL, Bash
Generative AI & LLMs: LangGraph, OpenAI Agents SDK, RAG, LLM Fine-Tuning, Prompt Engineering, DPO
Deep Learning & ML: PyTorch, Hugging Face Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn
Natural Language Processing: spaCy, NLTK, Sentiment Analysis, Text Processing
Data Engineering & Ops: Docker, AWS, FastAPI, Neo4j, BeautifulSoup, Jupyter

Selected Projects

HumorRank — LLM Evaluation & Ranking Framework | *Python, Evaluation* Leaderboard [Paper]
- Developed a scalable evaluation framework for ranking outputs from multiple LLMs on subjective generation tasks.
- Implemented pairwise comparison with probabilistic ranking using the Bradley-Terry model.
- Designed a candidate generation → evaluation → leaderboard workflow for scalable benchmarking.

HumorGen — Open-Source LLM Humor Generation Suite | *PyTorch* Website Hugging Face
- Developed HumorGen, releasing 14 open-source language models for controllable humor generation.
- Implemented alignment pipelines using supervised fine-tuning and Direct Preference Optimization (DPO).
- Designed the Cognitive Synergy framework for diverse humor generation.

MyTorch — Minimal Deep Learning Framework Built from Scratch | *Python*
- Implemented core tensor operations, automatic differentiation, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end model training while illustrating neural network internals.
- Built modular components for educational experimentation with deep learning systems.

Mental Health & Cyberbullying Detection System (AAAI Oral) | *Transformers* GitHub [arXiv]
- Built an end-to-end multi-label classification system for detecting mental health signals and cyberbullying from social media data.
- Combined classical ML methods with transformer embeddings for robust prediction on noisy datasets.
- Produced results leading to an Oral paper acceptance at AAAI-26 AIMedHealth.

AI Workflow Automation Agent | *Python, LangChain, FastAPI*
- Built an autonomous agent to process submissions, route tasks, and trigger approval workflows.
- Integrated API orchestration, notifications, and real-time validation pipelines.

Training & Conferences

Oral & Poster Presentation, AAAI Conference on Artificial Intelligence — Singapore
AAAI — *Jan. 2026*
- Presented my paper on ML for mental health and cyberbullying detection from social media, and engaged with academics and researchers at the conference.

Participant, Technical AI Safety Course — Online
Bluedot Impact — *May 2025 – Present*
- Training in technical AI safety, focusing on alignment approaches, mechanistic interpretability, and catastrophic risk mitigation.

Participant, Digital Humanities Summer School — Rwanda
DIGIHUM/CMU-Africa — *Jul. 2025*
- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

Founder & Organizer, CMU-Africa NLP Group
Carnegie Mellon University Africa — *Apr. 2025 – Present*
- Managing the group activities and overseeing research projects focused on language technologies.

President, CMU-Africa Research Club
Carnegie Mellon University — *Jan. 2025 – Dec. 2025*
- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation on campus.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AI for Medicine and Healthcare — *Nov. 2025 – Nov 2025*
- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- DataCamp Python Data Associate - View

Honors and Awards

- **Winner**, CLEF JOKER 2026 Task 4 — Best-performing humor generation system in English, Spanish, and French.
- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University